Lecture 21: Limits at Infinity

Goal: Understand the end behavior (asymptotic behavior) of functions.

Definition. We say:

$$\lim_{x\to\infty} f(x) = L$$

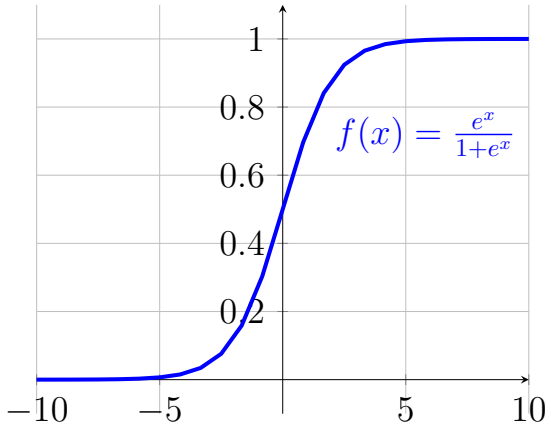
to mean $f(x)$ goes to L as x "goes to infinity" (i.e., x gets very, very large). Similarly,

$$\lim_{x\to-\infty} f(x) = L$$

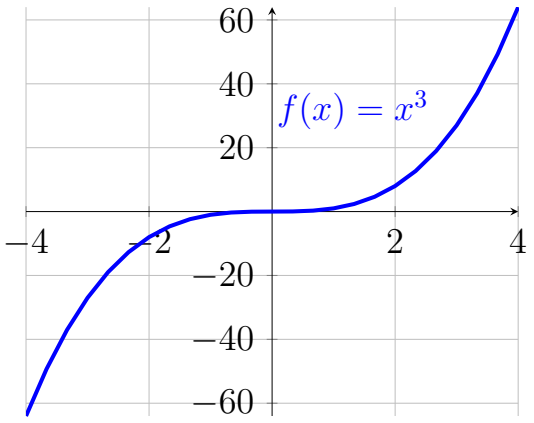
means $f(x)$ goes to L as x "goes to negative infinity" (i.e., x gets very, very "largely negative").

NOTE: A number being "small" typically refers to when it is close to 0. Hence, why I am using the phrase "largely negative," even though it is a bit awkward.

Example 1. For each part, determine $\lim_{x\to\infty} f(x)$ and $\lim_{x\to-\infty} f(x)$, where the graph of f is given.



$\lim_{x\to\infty} f(x) =$



$\lim_{x\to\infty} f(x) =$

$\lim_{x\to-\infty} f(x) =$

$\lim_{x\to-\infty} f(x) =$

Example 2. Determine the end behavior (limits at infinity) for the function $f(x) = \frac{1}{x}$

In general, for any positive integer n,

$$\lim_{x\to\infty} \frac{1}{x^n} = \lim_{x\to-\infty} \frac{1}{x^n} =$$

Example 3. Compute $\lim_{x\to\infty} \left(2 + \frac{3}{x^2}\right)$

Rational Functions

Main Strategy (Formally):

Example 4. For $f(x) = \dfrac{x^2 - 1}{2x^2 + 1}$, compute $\lim\limits_{x \to \infty} f(x)$ and $\lim\limits_{x \to -\infty} f(x)$.

Example 5. For $f(x) = \dfrac{x^2 + 2}{x^3 + x^2 - 1}$, compute $\lim\limits_{x \to \infty} f(x)$ and $\lim\limits_{x \to -\infty} f(x)$.

Main Strategy (Informally):

Example 6. For $f(x) = \dfrac{x^2 + x}{-x + 3}$, compute $\lim\limits_{x \to \infty} f(x)$ and $\lim\limits_{x \to -\infty} f(x)$.

In Summary: For a rational function:

$$f(x) = \frac{a(x)}{b(x)}$$

if $LT(a(x))$ and $LC(a(x))$ denote the leading term and leading coefficient of the polynomial p respectively, then,

$$\lim_{x \to \infty} f(x) = \lim_{x \to \infty} \frac{LT(a(x))}{LT(b(x))}$$

$$\lim_{x \to -\infty} f(x) = \lim_{x \to -\infty} \frac{LT(a(x))}{LT(b(x))}$$

In particular,

1. If $\deg(a(x)) = \deg(b(x))$, then:

$$\lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = \frac{LC(a(x))}{LC(b(x))}$$

2. If $\deg(a(x)) < \deg(b(x))$, then:

$$\lim_{x \to \infty} f(x) = \lim_{x \to -\infty} f(x) = 0$$

3. If $\deg(a(x)) > \deg(b(x))$, more work is needed (since it depends on whether $\deg(a(x)) - \deg(b(x))$ is even or odd as well as whether $\dfrac{LC(a(x))}{LC(b(x))}$ is positive or negative).

Horizontal and Vertical Asymptotes (HAs and VAs)

Definition. A function $f(x)$ has a **vertical asymptote at** $x = c$ if either $\lim_{x \to c^-} f(x)$ or $\lim_{x \to c^+} f(x)$ is NOT finite.

Definition. A function $f(x)$ has a **horizontal asymptote at** $y = L$ if either $\lim_{x \to -\infty} f(x) = L$ or $\lim_{x \to \infty} f(x) = L$.

Example 7. Find the locations of any Horizontal and Vertical Asymptotes for $f(x) = \dfrac{3x^2 + 5}{x^2 - 4}$

For a rational function, VAs occur when the denominator is _____, but the numerator is _____.

For a rational function $f(x)$, HAs occur when either $\lim_{x \to -\infty} f(x)$ or $\lim_{x \to \infty} f(x)$ is _____.

NOTE: Rational functions have at most one HA.

Slant (Oblique) Asymptotes

Recall Polynomial Long Division:

Theorem (Polynomial Division Algorithm)**.** For a rational function $\frac{a(x)}{b(x)}$ there are unique polynomials $q(x)$ (called the **quotient**) and $r(x)$ (called the **remainder**) such that:

$$\frac{a(x)}{b(x)} = q(x) + \frac{r(x)}{b(x)} \quad \text{AND} \quad \deg(r(x)) < \deg(b(x))$$

Example 8. Find the quotient and remainder when $2x^2 + 3x + 1$ is divided by $x - 1$.

Definition. The line $y = mx + b$ is a **slant (oblique) asymptote** of the graph of $f(x)$ if:

$$\lim_{x\to\infty}[f(x) - (mx + b)] = 0 \quad \text{OR} \quad \lim_{x\to-\infty}[f(x) - (mx + b)] = 0$$

Informally, this means the graph of $f(x)$ gets closer and closer to the line $y = mx + b$ as x goes to positive or negative infinity.

Theorem. The following are equivalent for a rational function $\dfrac{a(x)}{b(x)}$:

1. $\dfrac{a(x)}{b(x)}$ has a slant asymptote.
2. The quotient in polynomial long division is a degree 1 polynomial.
3. $\deg(a(x)) = \deg(b(x)) + 1$

If these hold, the _____ obtained from long division is a slant asymptote.

Example 9. Does the function $f(x) = \dfrac{2x^2 + 3x + 1}{x - 1}$ have a slant asymptote? If so, find it.

Example 10. Find all asymptotes (HAs, VAs, and slant) for the function $f(x) = \dfrac{2x^3 - 3x + 1}{x^2 - x - 6}$